Induced Subgraphs in Ramsey Graphs

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Examination Number: XXXXXXXX
Degree Course: MSc in Mathematical Sciences
Submission: Trinity Term 2026

Submitted in partial fulfilment of the requirements for the degree of Master in Mathematical Sciences (OMMS).

Word count:

Abstract

In this dissertation, we investigate the "random-like" features of Ramsey graphs, with an emphasis on the existence of induced subgraphs of many different sizes.

Contents

1 Introduction **3**
1.1 Notations and tools 6
1.2 The anticoncentration principle 7

2 Subgraphs with Many Degrees **9**

3 Subgraphs of Quadratically Many Sizes **13**
3.1 Pair diversity and richness 15
3.2 The main lemma and the construction 18
3.3 The construction 19
3.4 The double-exposure argument 22
3.5 Proof of the main lemma 26

4 The Erdős–McKay Conjecture **28**

5 Concluding Remarks **29**

A Supplementary Proofs **31**

1 Introduction

Classical Ramsey Theory asks for the minimum number of vertices required to guarantee the existence of a clique or an independent set in a graph. We refer to these maxima as *Ramsey numbers*.

Definition 1.1. The *Ramsey number* $R(s,t)$ is the minimum number N such that any graph on N vertices contains a clique of size s or an independent set of size t.

The existence of these numbers was proven by Ramsey [15] in 1930, but their quantitative behaviors remain largely unknown despite decades of intensive research. In particular, it is known that $R(3,3) = 6$ and $R(4,4) = 18$, but the exact values of $R(5,5)$ and beyond are yet to be determined. The cornerstone of Ramsey Theory due to Erdős and Szekeres in 1935 gives the following upper bound on $R(s,t)$:

Theorem 1.1 (Erdős–Szekeres [8])**.** *For $s,t \geq 2$,*

$$R(s,t) \leq \binom{s+t-2}{s-1}.$$

Following the fact that $\binom{2t-2}{t-1} < 2^{2t}$, we can see $R(t,t) < 4^t$. Very little progress had been made on this bound until the recent breakthrough of Campos, Griffiths, Morris, and Sahasrabudhe [4], who gave the first exponential improvement since the result of Erdős and Szekeres. On the other hand, the lower bound of ramsey numbers has been even more challenging to determine. A rather basic result which already gives the best known exponents is the famous random graph construction due to Erdős in 1947, marking one of the first applications of the probabilistic method in combinatorics. Before stating the result, we first define the *Erdős–Rényi random graph*.

Definition 1.2. For $n \geq 1$ and $p \in [0,1]$, the *random graph* $G_{n,p}$ is the graph on n vertices in which each of the $\binom{n}{2}$ edges is present independently with probability p.

Remark: For constant p, the random graph $G_{n,p}$ is dense in the sense that it has $\Theta(n^2)$ many edges almost surely.

Theorem 1.2 (Erdős [9])**.** *For $t \geq 3$,*

$$\sqrt{2}^t < R(t,t) < 4^t.$$

Proof. The upper bound follows from Theorem 1.1. For lower bound, set $N = \lfloor \sqrt{2}^t \rfloor$ and consider the random graph $G_{N,1/2}$. The expected number of cliques of size t is

$$\binom{N}{t} \cdot 2^{-\binom{t}{2}} < \frac{N^t}{t!} 2^{-\binom{t}{2}} \leq \frac{\sqrt{2}^{t^2}}{t!} 2^{-\binom{t}{2}} = \frac{\sqrt{2}^t}{t!} = \frac{\sqrt{2}^{t-3}}{t(t-1)\cdots 4} \cdot \frac{\sqrt{2}^3}{3!} \leq \frac{\sqrt{2}^3}{3!} < \frac{1}{2}.$$

Similarly, the expected number of independent sets of size t is less than $1/2$. It now follows that there exists a graph on N vertices containing neither a clique nor an independent set of size t. □

The above strategy may be optimized to give a lower bound of $\frac{t}{e\sqrt{2}}\sqrt{2}^t$, which has been astonishingly resilient to improvement. The only noteworthy improvement since 1947 is achieved in 1977 by Spencer [17], who used the Lovász local lemma to raise the bound to $(1-o(1))\frac{t\sqrt{2}}{e}\sqrt{2}^t$, an improvement by a factor of 2. For further background on Ramsey theory, see Conlon, Fox, and Sudakov [6].

From Theorem 1.2, we can see that any graph on n vertices contains a clique or independent set of size at least $\frac{1}{2}\log_2 n$, whereas there exists a graph on n vertices containing neither a clique nor an independent set of size $2\log_2 n$. Apart from determining the bounds of Ramsey numbers, the search for explicit graph constructions with logarithmically small cliques and independent sets has also been a major open problem in Ramsey Theory. Despite decades of intensive research, all known constructions still rely on some randomness. This thus motivates the question of whether graphs with small cliques and independent sets inherently resemble dense random graphs like $G_{n,1/2}$; we refer to such graphs as *Ramsey graphs*.

Definition 1.3. Let $C > 0$ be a constant. A *C-Ramsey graph* is a graph with no clique or independent set of size greater than $C\log n$.

It is useful to also define a shorthand for cliques and independent sets.

Definition 1.4. A set $S \subseteq V(G)$ is *homogeneous* in a graph G if it induces a clique or an independent set in G. We denote the size of the largest homogeneous set in G by $\hom(G)$.

The belief that Ramsey graphs and random graphs are intimately related is supported a number of theorems showing that certain "richness" properties of random graphs also hold for Ramsey graphs. The first result that cemented this connection is due to Erdős and Szemerédi, showing that Ramsey graphs are also dense.

Theorem 1.3 (Erdős–Szemerédi [9])**.** *Let G be a graph on n vertices with edge density $d \le 1/2$. Then G contains a homogeneous set of size greater than $K = \frac{c}{d\log(1/d)}\log n$, for some constant $c > 0$ and sufficiently large n.*

Proof. Suppose not. Pick $c < 1/32$. Since G contains $d\binom{n}{2}$ edges, there exists $m \ge n/2$ vertices in G with degree at most dn, by the pigeonhole principle. We work with the graph G' induced by these m vertices. Let I be the largest independent set in G' and let $k = |I|$. Since $\deg(v) \le dn$ for all $v \in I$, we have $\sum_{v\in I}\deg(v) \le dnk \le 2dmk$. Put $s = 8dk$. Again by the pigeonhole principle, there are at least $m/2 - k$ vertices $v \in V(G')\setminus I$ with $\deg_I(v) < s$. Let J be the set of these vertices. Consider the family $\mathcal{F}$ of subsets of I with size less than s. Partition J by each vertex's neighbourhood in I, and note that each resulting class corresponds to an element of $\mathcal{F}$. By the standard bounds $\binom{x}{r} \le (ex/r)^r$ and $1 + x \le e^x$,

$$|\mathcal{F}| \le s\binom{k}{s-1} \le s\left(\frac{ek}{s-1}\right)^{s-1} = s\left(\frac{s}{s-1}\right)^{s-1}\left(\frac{ek}{s}\right)^{s-1} \le se\left(\frac{ek}{s}\right)^{s-1} = \frac{s^2}{k}\left(\frac{ek}{s}\right)^{s}.$$

By assumption $|I| = k < K$, and so

$$s < 8dK = \frac{8c\log n}{\log(1/d)} = O(\log n).$$

Since $s/k = 8d$, it follows that

$$|\mathcal{F}| \leq \frac{s^2}{k}\left(\frac{ek}{s}\right)^s \leq O(\log n)\cdot\left(\frac{e}{8}\cdot\frac{1}{d}\right)^s \leq O(\log n)\cdot\left(\frac{1}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = O(\log n)\cdot n^{8c} < n^{1/3}.$$

But then

$$\frac{|J|}{|\mathcal{F}|} \geq \frac{m/2-k}{n^{1/3}} \geq \sqrt{n},$$

so by the pigeonhole principle, there exists a class of vertices Z in J of size at least $\sqrt{n}$. Let $G'' = G[Z]$. Note that G'' does not contain an independent set of size s, otherwise exchange it with Y in I creates a larger independent set in G', contradicting the maximality of I. Let t be the size of the largest clique in G'', and note that $t < K$ by assumption. The Erdős-Szekeres and the standard binomial bounds give

$$\sqrt{n} \leq |Z| \leq \binom{s+K-2}{s-1} \leq \binom{s+K}{s} \leq \binom{8dK+K}{8dK} \leq \left(\frac{e(8d+1)}{8d}\right)^s \leq \left(\frac{2}{d}\right)^s,$$

where the last inequality holds because $e(1+8d)/8 \leq 5e/8 \leq 2$ for $d \leq 1/2$. But then since $d \leq 1/2$ and $c < 1/32$,

$$\left(\frac{2}{d}\right)^s \leq \left(\frac{2}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = n^{8c\cdot\frac{\log(2/d)}{\log(1/d)}} = n^{8c\left(1+\frac{\log 2}{\log(1/d)}\right)} \leq n^{16c} < \sqrt{n},$$

contradiction. □

Recall that the Erdős–Szemerédi theorem gives the bound $\hom(G) \geq \frac{1}{2}\log_2 n$, regardless of the edge density. Theorem 1.3 improves this when the density is small. For example, if $d = 1/n^\varepsilon$ for some $\varepsilon > 0$, then $\hom(G) = \Omega\left(\frac{n^\varepsilon}{\log n^\varepsilon}\log n\right)$, vastly larger than $\frac{1}{2}\log_2 n$. Intuitively, a sparse graph should contain a large independent set, and the theorem makes this notion precise.

If G is C-Ramsey, then applying Theorem 1.3 to whichever G or $\overline{G}$ has density $d \leq 1/2$ yields $c/(d\log(1/d)) \leq C$, forcing d to be bounded away from 0. By the same argument applied to $\overline{G}$ forces d to be bounded away from 1. We thus have the following corollary.

Corollary 1.4. *For any C-Ramsey graph G, there exists $\delta = \delta(C) \in (0, 1/2)$ such that the edge density of G lies in the interval $[\delta, 1-\delta]$.*

Due to the rich variety of induced subgraphs in dense random graphs, a popular angle to study the pseudorandom properties of Ramsey graphs is to consider their induced subgraphs. Some notable results are a theorem of Shelah [16] which asserts

that every n-vertex Ramsey graph contains $2^{\Omega(n)}$ non-isomorphic induced subgraphs, and a theorem of Prömel and Rödl [14] which shows the existence of $\delta = \delta(C) > 0$ such that every C-Ramsey graph on n vertices contains an induced copy of all graphs on at most $\delta \log n$ vertices.

It is also interesting to look at the edge statistics of Ramsey graphs. For a graph G, let

$$\Phi(G) \coloneqq \{e(H) : H \text{ is an induced subgraph of } G\}$$

Erdős and McKay [7] conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G).$$

It was proven for Random graphs by Calkin, Frieze, and Mckay [3], but the conjecture for Ramsey graphs remained open for decades and only started to show significant progress in recent years. In particular, a chain of results were sparked after the work of Narayanan, Sahasrabudhe, and Tomon [13] establishing that $|\Phi(G)| = n^{2-o(1)}$ for Ramsey graph G on n vertices. Their proof builds on some ideas from the work of Bukh and Sudakov [2] which showed that every Ramsey graph contains a linear sized subgraph with many distinct degrees, a feature that dense random graphs also exhibit. Kwan and Sudakov [11] later improved the bound to $|\Phi(G)| = \Omega(n^2)$ by refining the argument of Narayanan, Sahasrabudhe, and Tomon. Eventually, Kwan, Sah, Sauermann, and Sawhney [12] closed the conjecture with a fairly involved approach using tools from Fourier analysis, random matrix theory, and the theory of Boolean functions.

This dissertation will start by studying the result of Bukh and Sudakov [2] on large induced subgraphs in Ramsey graphs with many distinct degrees. This work not only tightens the connection between Ramsey graphs and random graphs, but also introduces a key feature of Ramsey graphs that unlocks the subsequent $|\Phi(G)| = \Omega(n^2)$ result by Kwan and Sudakov, which we will present in the following chapter. Lastly, we go through the broad strokes of the proof to the Erdős–McKay conjecture by Kwan, Sah, Sauermann, and Sawhney [12], touching on the core ideas of their approach.

1.1 Notations and tools

We call $e(G)$ the *size* of a graph G. We omit all floor and ceiling signs when they are not crucial for simplicity.

Proposition 1.5 (Turán's theorem)**.** *Every graph G with n vertices and m edges contains an independent set of size at least $n^2/(2m+n)$.*

Proposition 1.6 (Hoeffding's inequality)**.** *Let $X_1, X_2, \ldots, X_n$ be independent real-valued random variables with $0 \le X_i \le 1$ for each i, and write $X = \sum_{i=1}^n X_i$. Then for $t \ge 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \ge t) \le 2e^{-2t^2/n}.$$

Proposition 1.7. *Let X denote the number of successes in D draws, without replacement, from a population of size N which contains M successes. Then $\mathbb{E}[X] = DM/N$ and for $t \geq 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq 2e^{-2t^2 D}.$$

Proof. □

Definition 1.5 (Littlewood-Offord type random variable)**.** A random variable X is of *(n,p)-Littlewood-Offord type* if it is of the form $X = a_1\xi_1 + a_2\xi_2 + \cdots + a_n\xi_n + C$, where $a_1, a_2, \ldots, a_n \in \mathbb{Z} \setminus \{0\}$ and $C \in \mathbb{Z}$ are fixed and $\xi_1, \xi_2, \ldots, \xi_n$ are independent Bernoulli random variables with parameter p.

Proposition 1.8 (Littlewood-Offord theorem)**.** *Let $p > 0$ be constant and X be a random variable of (n,p)-Littlewood-Offord type. Then for any $t \in \mathbb{Z}$,*

$$\mathbb{P}[X = t] = O(1/\sqrt{n}).$$

1.2 The anticoncentration principle

Before diving into the core results, we introduce a key probabilistic mechanism which serves as a unifying thread throughout the dissertation. In the following chapters, we will repeatedly sample a random vertex set $U \subseteq V(G)$ and study the induced subgraph $G[U]$. One natural question to ask in this setting is how likely are two vertices $x, y \in V(G)$ to have the same degree in $G[U]$?

Observe that $\deg_U(x) - \deg_U(y)$ counts only vertices in the symmetric difference $D = N_G(x)\Delta N_G(y)$. Writing $A = N(x) \setminus N(y)$ and $B = N(y) \setminus N(x)$, we have

$$\deg_U(x) - \deg_U(y) = |U \cap A| - |U \cap B|, \tag{1}$$

x and y have the same degree in $G[U]$ precisely when $|U \cap A| = |U \cap B|$.

Suppose first that U is formed by including each vertex independently with probability p. Then each vertex $v \in A$ independently contributes $+1$ to the difference $|U \cap A| - |U \cap B|$ with probability p, while each vertex $v \in B$ independently contributes -1 to the difference with probability p. The difference is thus a sum of $|D| = |A| + |B|$ independent bounded random variables. When $|D|$ is large, say $|D| \geq \varepsilon n$, this sum has standard deviation $\Theta(\sqrt{n})$. Since $|U \cap A| - |U \cap B|$ is integer-valued, we can show the probability that it equals any particular integer is at most $O(1/\sqrt{n})$ via the one-dimensional Erdős-Littlewood-Offord theorem. Thus we conclude that $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$. We will formalize this with greater detail in Chapter 3.

If instead U is a uniformly random subset of a fixed size m, the vertices of U are no longer independent. Nevertheless, conditioning on $k = |U \cap D|$, we may observe that, given k, the split of these k elements between A and B follows a hypergeometric distribution. With the same assumption that $|D|$ is large, Stirling's approximation

applied to the hypergeometric probability then yields the same $O(1/\sqrt{n})$ bound. We will carry out this calculation in detail in Chapter 2.

An intuition to the bound $O(1/\sqrt{n})$ is that, in either sampling method, the difference $\deg_U(x) - \deg_U(y)$ is approximately normally distributed with standard deviation $\Theta(\sqrt{n})$. In the independent case, this follows from the central limit theorem applied to a sum of independent bounded random variables. In the fixed-size case, it follows from the normal approximation to the hypergeometric distribution. Since the probability density function $\phi(x)$ of a normal distribution with standard deviation σ is at most $1/(\sigma\sqrt{2\pi})$ for all x, the probability that $\deg_U(x) - \deg_U(y)$ equals any particular value is $O(1/\sigma) = O(1/\sqrt{n})$. The theorems in the following chapters will rely on this principle. In Chapter 2, it bounds the probability that two vertices share a degree in a random induced subgraph, which leads to $\Omega(\sqrt{n})$ distinct degrees. In Chapter 3, the same principle bounds the probability that two vertices have equal degree in a random induced subgraph, which is a key step in producing $\Omega(n^2)$ distinct subgraph sizes. The resolution of the Erdős–McKay conjecture in Chapter 4 requires a substantially more powerful approach via Fourier analysis, but the underlying idea remains: when a random variable is approximately normal with large standard deviation, its point probabilities are small.

2 Subgraphs with Many Degrees

In this chapter, we present the following result of Bukh and Sudakov.

Theorem 2.1 (Bukh–Sudakov [2])**.** *Every Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees.*

This was originally conjectured by Erdős, Faudree, and Sós [7] in light of the observation that the random graph $G_{n,1/2}$ exhibits such a property almost surely, but the $\Omega(\sqrt{n})$ bound is in fact not optimal for both Ramsey graphs and $G_{n,1/2}$. In particular, an upper bound of $O(n^{2/3})$ is noted in [2], which was later proved to be the correct order for Ramsey graphs by Jenssen, Keevash, Long, and Yepremyan [10] via a continuous relaxation approach. The corresponding order for $G_{n,1/2}$ was also matched in an unpublished work of Conlon, Morris, Samotij, and Saxton [5], further reinforcing the case that Ramsey graphs resemble random graphs in various respects. We will, however, focus on the original result of Bukh and Sudakov, as it introduces a key characteristic of Ramsey graphs that will be insightful in the following chapter.

Definition 2.1. An n-vertex graph G is (ε, δ)-*diverse* if for $x \in V(G)$, we have $|N(x)\Delta N(y)| \geq \varepsilon n$ for all but at most n^δ vertices $y \in V(G)$.

The notion of diversity captures the idea that the neighbourhoods of most pairs of vertices differ significantly. In particular, (ε, δ)-diversity directly conditions our assumption that $|N(x)\Delta N(y)| = D \geq \varepsilon n$ in the discussion of the anticoncentration principle in Section 1.2. Hence, following our discussion in Section 1.2, we see that (ε, δ)-diversity implies that $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$ for all but at most n^δ pairs of vertices $x, y \in V(G)$, from which we use Jensen's inequality to conclude that there are many distinct degrees in a diverse graph. The following lemma formalizes this observation and reduces Theorem 2.1 to showing every Ramsey graph contains a diverse subgraph of linear size.

Lemma 2.2. *Let $\delta \in (0, 1/2)$ and let G be an (ε, δ)-diverse graph on n vertices for large enough n. Then for $n/4 \leq m \leq 3n/4$, there is a subgraph on m vertices containing vertices of $\sqrt{\varepsilon n}/320$ distinct degrees.*

Proof. Fix m as above. Let U be chosen uniformly at random among all subsets of $V(G)$ of size m. Let $H = G[U]$. It suffices to show that with positive probability, H contains $\Omega(\sqrt{n})$ vertices of distinct degrees.

For distinct $x, y \in V(G)$, define $\mathbb{1}_{xy}$ to be the indicator for the event that $x, y \in V(H)$ and $\deg_H(x) = \deg_H(y)$. Then $\mathbb{E}[\mathbb{1}_{xy}] \leq \mathbb{P}[\deg_U(x) = \deg_U(y)]$. Put $D = N_G(x)\Delta N_G(y)$, and $k = |D \cap U|$. Write $A = N_G(x) \setminus N_G(y)$ and $B = N_G(y) \setminus N_G(x)$. Then

$$\mathbb{P}[\deg_U(x) = \deg_U(y)] = \mathbb{P}\left[|A \cap U| = |B \cap U|\right].$$

Suppose $|D| \geq \varepsilon n$. Since $\frac{|D|}{4} \leq \frac{m|D|}{n} \leq \frac{3|D|}{4}$, Proposition 1.7 yields

$$\mathbb{P}\left[k < \frac{|D|}{8} \text{ or } k > \frac{7|D|}{8}\right] \leq \mathbb{P}\left[|k - \mathbb{E}[k]| > \frac{|D|}{8}\right] \leq e^{-\Theta(|D|)} = e^{-\Theta(n)} = o(n^{-1/2}).$$

On the other hand, recall that U is a uniformly random subset of a fixed size m. Since every k-subset of D is equally likely to be $U \cap D$ by symmetry, conditioning on $|U \cap D| = k$, the k elements of U that land in D form a uniformly random subset of D of size k. It follows that when conditioning on k, the event $|A \cap U| = |B \cap U| = k/2$ follows the hypergeometric distribution $\mathrm{Hyp}(|D|, |A|, k)$. Thus,

$$\mathbb{P}\left[\deg_U(x) = \deg_U(y) \,\middle|\, \frac{|D|}{8} \leq k \leq \frac{7|D|}{8}\right] = \frac{\binom{|A|}{k/2}\binom{|B|}{k/2}}{\binom{|D|}{k}}.$$

Since $|A| + |B| = |D|$, by the log-concavity of the binomial coefficients we have $\binom{|A|}{k/2}\binom{|B|}{k/2} \leq \binom{|D|/2}{k/2}^2$. It now follows from Stirling's formula and $k(|D| - k) \geq \frac{7}{64}|D|^2$ that the above probability does not exceed

$$\frac{\binom{|D|/2}{k/2}^2}{\binom{|D|}{k}} = (1 + o(1))\sqrt{\frac{2|D|}{\pi k(|D| - k)}} \leq \frac{3}{\sqrt{|D|}} \leq \frac{3}{\sqrt{\varepsilon n}}. \tag{2}$$

Thus if $|N_G(x) \Delta N_G(y)| \geq \varepsilon n$ then $\mathbb{E}[\mathbb{1}_{xy}] \leq 3/\sqrt{\varepsilon n} + o(n^{-1/2}) \leq 4/\sqrt{\varepsilon n}$.

Let $I = \sum_{x,y \in V(G)} \mathbb{1}_{xy}$. By definition of an (ε, δ)-diverse graph, we have $\mathbb{E}[\mathbb{1}_{xy}] \leq 4/\sqrt{\varepsilon n}$ for all but at most $n^{1+\delta} < n^{3/2}$ pairs $x, y \in V(G)$. Thus,

$$\mathbb{E}[I] \leq n^{1+\delta} + \frac{4}{\sqrt{\varepsilon n}} \cdot n^2 \leq \frac{5}{\sqrt{\varepsilon}} n^{3/2},$$

and so there is a choice for U such that $I \leq 5n^{3/2}/\sqrt{\varepsilon}$. Now let r be the number of vertices of distinct degrees in H. Let n_k be the number of vertices in H with degree k and note that $\sum_k n_k = m$. By Jensen's inequality

$$5n^{3/2}/\sqrt{\varepsilon} \geq I = \sum_k \binom{n_k}{2} \geq r\binom{\frac{1}{r}\sum_k n_k}{2} \geq \frac{m^2}{4r} \geq \frac{n^2}{64r}.$$

Rearranging yields the lemma. □

Remark. The hypergeometric distribution $\mathrm{Hyp}(|D|, |A|, k)$ has variance

$$\sigma^2 = \frac{k|A||B|(|D| - k)}{|D|^2(D - 1)} \leq \frac{k(|D| - k)}{4(|D| - 1)},$$

where inequality follows from AM-GM. When k is not the tail end of the distribution, we have $\sigma = \Theta(\sqrt{D}) = \Theta(\sqrt{n})$, and the Stirling bound in (2) is asymptotically $1/(\sigma\sqrt{2\pi})$ as $n \to \infty$, which is exactly the maximum density of a normal distribution

with standard deviation σ. This confirms the intuition from Section 1.2 that the degree difference is approximately normally distributed so its point probabilities are $O(1/\sigma) = O(1/\sqrt{n})$.

As mentioned above, it remains to show that every Ramsey graph contains a diverse subgraph of linear size.

Lemma 2.3. *For $\delta \in (0,1)$, there exists $\varepsilon = \varepsilon(C,\delta) > 0$ such that every C-Ramsey graph G on n vertices contains an (ε,δ)-diverse subgraph on $\Omega(n)$ vertices.*

Proof. Suppose not. Let $K = K(C)$ be a large constant to be determined. Let $c = 8^{-K}$ and $\varepsilon = 1/K$. We inductively construct a sequence of disjoint vertex sets $S_1, S_2, \ldots, S_K$ of size $(cn)^\delta$, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 8\varepsilon$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 8\varepsilon$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G_0 \supset G_1 \supset \cdots \supset G_K$, such that $|G_i| \geq |G_{i-1}|/8$ and $S_i \subseteq V(G_{i-1})$ for all i.

We first show that this construction suffices. By working with $\overline{G}$ instead of G if necessary, we may assume that $r \geq K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^r S_{i_k}$. Since each S_{i_k} has size $(cn)^\delta$ and $\varepsilon = 1/K$,

$$d(S) \leq \frac{\binom{(cn)^\delta}{2}}{\binom{r(cn)^\delta}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \leq \frac{1}{r} + 8\varepsilon \leq \frac{10}{K}.$$

Thus Theorem 1.3 yields a homogeneous set A in S of size

$$|A| \geq \frac{\gamma K}{10\log(K/10)} \log|S| = \frac{\gamma\delta K}{10\log(K/10)} \log cn = \frac{\gamma\delta K}{10\log(K/10)} \log n + o(\log n),$$

for some absolute constant $\gamma > 0$ given by the theorem. But then $\frac{\gamma\delta K}{10\log(K/10)} > C$ for sufficiently large K, and so $|A| > C\log n$, contradiction.

We now construct the sets S_i and G_i inductively. Suppose $G_1, \ldots, G_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. By induction, $|G_{i-1}| \geq 8^{1-i}|G| \geq cn$. Note that G_{i-1} is not (ε,δ)-diverse by assumption, so there exists a vertex $x \in V(G_{i-1})$ and a set $S_i \subseteq G_{i-1}$ of size $(cn)^\delta$ such that $|N_{G_{i-1}}(x) \Delta N_{G_{i-1}}(y)| < \varepsilon|G_{i-1}|$ for $y \in S_i$. Let $T_i = V(G_{i-1}) \setminus S_i$. Since $(cn)^\delta \leq cn/2$ for large enough n, we have $|T_i| \geq |G_{i-1}|/2$.

We consider two cases. In the first case, suppose $|\overline{N_{T_i}}(x)| \geq |T_i|/2$. For $y \in S_i$,

$$|N_{T_i}(y) \cap \overline{N}_{T_i}(x)| \leq |N_{T_i}(x) \Delta N_{T_i}(y)| < \varepsilon|V(G_{i-1})|,$$

and so

$$d(y, \overline{N}_{T_i}(x)) = \frac{|N_{T_i}(y) \cap \overline{N}_{T_i}(x)|}{|N_{T_i}(x)|} < \frac{\varepsilon|V(G_{i-1})|}{|T_i|/2} \leq 4\varepsilon.$$

Partition $\overline{N}_{T_i}(x)$ into $V_i \sqcup Z_i$, where $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) < 8\varepsilon\}$. We double count $e(\overline{N}_{T_i}(x), S_i)/|S_i|$. Since $d(v, S_i) \geq 8\varepsilon$ for $v \in Z_i$,

$$\frac{e(\overline{N}_{T_i}(x), S_i)}{|S_i|} = \sum_{v \in \overline{N}_{T_i}(x)} \frac{e(v, S_i)}{|S_i|} = \sum_{v \in \overline{N}_{T_i}(x)} d(v, S_i) \geq \sum_{v \in Z_i} d(v, S_i) \geq 8\varepsilon |Z_i|.$$

On the other hand, since we showed $d(y, \overline{N}_{T_i}(x)) < 4\varepsilon$ for $y \in S_i$,

$$\frac{e(\overline{N}_{T_i}(x), S_i)}{|S_i|} = \frac{1}{|S_i|} \sum_{y \in S_i} e(y, \overline{N}_{T_i}(x)) = \frac{|\overline{N}_{T_i}(x)|}{|S_i|} \sum_{y \in S_i} d(y, \overline{N}_{T_i}(x)) \leq 4\varepsilon |\overline{N}_{T_i}(x)|.$$

Combining both inequalities yields $|Z_i| \leq |\overline{N}_{T_i}(x)|/2$. It follows that

$$|V_i| \geq \frac{|\overline{N}_{T_i}(x)|}{2} \geq \frac{|T_i|}{4} \geq \frac{|V(G_{i-1})|}{8}.$$

Recall that $d(S_i, U) < 8\varepsilon$ for all $U \subseteq V_i$. Setting $G_i = G[V_i]$ thus satisfy $|V(G_i)| \geq |V(G_{i-1})|/8$ and condition (i) for S_i.

The case for $|N_{T_i}(x)| \geq |T_i|/2$ follows the same argument. Since any vertex in $N_{T_i}(x)$ that is not a neighbor of y lies in $N_{T_i}(x) \setminus N_{T_i}(y) \subseteq N_{T_i}(x) \Delta N_{T_i}(y)$, we have

$$|N_{T_i}(x) \cap N_{T_i}(y)| \geq |N_{T_i}(x)| - |N_{T_i}(x) \Delta N_{T_i}(y)| < |N_{T_i}(x)| - \varepsilon |V(G_{i-1})|,$$

and so

$$d(y, N_{T_i}(x)) = \frac{|N_{T_i}(y) \cap N_{T_i}(x)|}{|N_{T_i}(x)|} > \frac{|N_{T_i}(x)| - \varepsilon |V(G_{i-1})|}{|N_{T_i}(x)|} \geq 1 - \frac{\varepsilon |V(G_{i-1})|}{|T_i|/2} \geq 1 - 4\varepsilon.$$

Partition $N_{T_i}(x)$ into $V_i \sqcup Z_i$, where $V_i = \{v \in N_{T_i}(x) : d(v, S_i) > 1 - 8\varepsilon\}$. Then for $v \in Z_i$, we have $1 - d(v, S_i) \geq 8\varepsilon$, so

$$8\varepsilon |Z_i| \leq \sum_{v \in Z_i} (1 - d(v, S_i)) \leq \sum_{v \in N_{T_i}(x)} (1 - d(v, S_i)).$$

On the other hand, since we showed $d(y, N_{T_i}(x)) > 1 - 4\varepsilon$ for $y \in S_i$.

$$\frac{e(N_{T_i}(x), S_i)}{|S_i|} = \frac{|N_{T_i}(x)|}{|S_i|} \sum_{y \in S_i} d(y, N_{T_i}(x)) \geq (1 - 4\varepsilon) |N_{T_i}(x)|,$$

But then

$$\sum_{v \in N_{T_i}(x)} (1 - d(v, S_i)) = |N_{T_i}(x)| - \sum_{v \in N_{T_i}(x)} d(v, S_i) \geq 4\varepsilon |N_{T_i}(x)|.$$

Combining both inequalities on $\sum (1 - d(v, S_i))$ yields $|Z_i| \leq |N_{T_i}(x)|/2$, and so $|V_i| \geq\geq |V(G_{i-1})|/8$. Note that $d(S_i, U) > 1 - 8\varepsilon$ for all $U \subseteq V_i$. Setting $G_i = G[V_i]$ thus satisfies $|V(G_i)| \geq |V(G_{i-1})|/8$ and condition (ii) for S_i. This completes the induction. $\square$

Theorem 2.1 follows immediately from Lemmas 2.2 and 2.3.

3 Subgraphs of Quadratically Many Sizes

Recall Erdős and Mckay conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G) \coloneqq \{e(H) : H \text{ is an induced subgraph of } G\},$$

That is, for any $1 \leq t \leq \delta n^2$, there exists an induced subgraph with exactly t edges. In [3], the conjecture was proven for random graphs, and so it is natural to ask whether Ramsey graphs also exhibit this property. The conjecture was widely open for decades. An early major result was due to Alon, Krivelevich and Sudakov in 2003 [1] which proved the conjecture with n^δ in place of δn^2, but little progress was made afterwards. Note that Erdős and McKay's conjecture implies that $|\Phi(G)| = \Omega(n^2)$, which is best possible asymptotically as $|\Phi(G)| \leq \binom{n}{2}$. Thus a natural relaxation of the conjecture is to ask whether Ramsey graphs have quadratically many distinct sizes of induced subgraphs.

In 2017, Narayanan, Sahasrabudhe, and Tomon [13] proved a near optimal bound of $|\Phi(G)| = n^{2-o(1)}$ by building on the notion of diversity from Chapter 2. The desired quadratic bound was achieved by Kwan and Sudakov [11] a few years later, and this chapter is dedicated to presenting their proof.

Theorem 3.1 (Kwan–Sudakov [11])**.** *Every C-Ramsey graph on n vertices contains induced subgraphs of $\Omega(n^2)$ distinct sizes.*

Bukh and Sudakov observed that if a graph G has k distinct degrees, then the induced subgraphs $G - \{v\}$ for $v \in V(G)$ have at least k different sizes. Thus Theorem 2.1 provides a simple bound of $|\Phi(G)| = \Omega(\sqrt{n})$ for any Ramsey graph G. Two ideas may be used to improve this construction. First, we can apply the observation to many different induced subgraphs rather than merely G itself. Second, instead of deleting only one vertex, we may add and remove various combinations of vertices to obtain many more distinct sizes. These ideas form the backbone of both the result of Narayanan, Sahasrabudhe, and Tomon and the proof of Theorem 3.1. We outline their shared strategy before describing what separates them.

The basic approach, introduced in [13] and inherited by [11], proceeds in two stages. In the first stage, we pick a target m and find two disjoint vertex sets U and W, such that U is a linear-sized random set with $e(U) \approx m$, and W is a small set of size $O(\sqrt{n})$ which will be used to augment U. In particular, W is selected from a candidate set $W_0 \subseteq V(G) \setminus U$ whose degrees all lie in a narrow interval centered around some $d = \Theta(n)$. In the second stage, we show that the subsets $Z \subseteq W$ produce $\Omega(n^{3/2})$ distinct values of $e(U \cup Z)$. Since $|W| = O(\sqrt{n})$ and each vertex $w \in W$ has at most n neighbours in $U \cup W$, we have $|e(U \cup W) - e(U)| = O(n^{3/2})$, which implies that all $\Omega(n^{3/2})$ values lie in an interval of length $O(n^{3/2})$ centred at $e(U)$. Applying this to $\Omega(\sqrt{n})$ well-separated targets m yields the full $\Omega(n^2)$ count.

Kwan and Sudakov implement the second stage as follows. The augment set W is composed of three disjoint parts S, T, X of size $\Theta(\sqrt{n})$, each with their own role.

Each element in S, T, and X has $\Theta(n)$ edges into U, and the degrees from S into U are uniformly smaller than those from T by a gap of $\Omega(\sqrt{n})$. By varying the number k of elements of S included in Z, we shift $e(U \cup Z)$ at the coarsest scale, where each additional element changes the edge count by $\Theta(n)$. By exchanging an element of S in Z with an element of T, we shift the edge count by $\Theta(\sqrt{n})$, as the two elements have the same number of edges into U up to a difference of $\Theta(\sqrt{n})$. Together, these two operations generate $\Theta(n)$ distinct values of $e(U \cup Z)$, roughly evenly spaced at intervals of $\Theta(\sqrt{n})$. To fill in the remaining $\Theta(\sqrt{n})$ gap between consecutive values, we include exactly one element of X in Z. The key requirement is that, for each choice of $Z \cap (S \cup T)$, the elements of X have $\Theta(\sqrt{n})$ distinct degrees into $U \cup Z$, contributing $\Theta(\sqrt{n})$ further distinct values. In total, we obtain $\Theta(n^{3/2})$ distinct values of $e(U \cup Z)$.

Achieving this last requirement that X provides $\Theta(\sqrt{n})$ distinct degrees into $U \cup Z$ is the main technical challenge. It is here that the proofs of Narayanan, Sahasrabudhe, and Tomon and of Kwan and Sudakov diverge. In [13], the second stage is handled differently. Narayanan, Sahasrabudhe, and Tomon reduced the problem to a weighted-graph question on $G[W]$ by assgining weight $\omega(w) = d_U(w) - d$ to each vertex, and a recursive probabilistic argument shows that the weighted Ramsey graph $G[W]$ has $|W|^{3-o(1)}$ distinct weighted subgraph sizes. This approach already yields $|\Phi(G)| = n^{2-o(1)}$, but the $n^{o(1)}$ loss turns out to be inherent to the use of diversity.

As discussed in Section 1.2, given a uniformly random set U and vertices $x, y \in V(G)$, we have

$$\mathbb{P}[d_U(x) = d_U(y)] = O(1/\sqrt{n})$$

whenever $|N(x)\Delta N(y)| = \Omega(n)$. We form a "degree-collision" graph H on the candidate set W_0 by joining x and y whenever $d_U(x) = d_U(y)$, and note that every independent set in H corresponds to a set of vertices with distinct degrees into U. Recall that the structural condition guaranteeing $|N(x)\Delta N(y)| = \Omega(n)$ is (ε, δ)-diversity, which ensures this for all but at most n^δ exceptional vertices y per vertex x. The expected number of edges in H is thus

$$\mathbb{E}[e(H)] = O(|W_0|^2/\sqrt{n} + |W_0| \cdot n^\delta),$$

where the first term accounts for diverse pairs, each contributing $O(1/\sqrt{n})$, and the second term accounts for pairs involving an exceptional vertex, each contributing at most 1. By Turán's theorem (Proposition 1.5), H contains an independent set of size at least $|W_0|^2/(2\mathbb{E}[e(H)] + |W_0|)$. For this to yield $\Omega(\sqrt{n})$ vertices with distinct degrees, the first term must dominate the second, requiring $|W_0| = \Omega(n^{1/2+\delta})$. But W_0 is constructed via the pigeonhole principle as a set of vertices whose degrees lie in an interval of length $O(\sqrt{n})$, and since there are n vertices and $O(\sqrt{n})$ such intervals, we can only guarantee $|W_0| = \Theta(\sqrt{n})$. The collision bound then gives an independent set of size $\Omega(n^{1/2-\delta})$. Since Lemma 2.2 allows δ to be taken arbitrarily small, this yields $|\Phi(G)| = n^{2-o(1)}$, which does not reach the targeted bound $\Omega(n^2)$.

The innovation from Kwan and Sudakov is to allow S, T, X to be sets of disjoint *pairs* of vertices rather than single vertices, treating each pair as a single unit. We write a variable in bold (e.g. $\mathbf{x}$) to indicate that it may refer to either a vertex or a pair, and similarly for sets of such objects (e.g. $\mathbf{S}$, $\mathbf{X}$). When $\mathbf{x} = \{x_1, x_2\}$ is a pair, we write $d(\mathbf{x}) = d(x_1) + d(x_2)$ and define $N(\mathbf{x})$ as the multiset union of $N(x_1)$ and $N(x_2)$. There are $\binom{n}{2} = \Theta(n^2)$ vertex pairs in G, but only $O(n)$ possible values of the pair-degree $d(\{v_1, v_2\}) = d(v_1) + d(v_2)$ for $\{v_1, v_2\} \in \binom{V(G)}{2}$. By the pigeonhole principle, some pair-degree value is shared by at least $\Theta(n)$ disjoint pairs. This is far larger than $n^{1/2+\delta}$ for any small $\delta > 0$, and so the n^δ exceptional-pair term becomes negligible in the collision count. The price is that we need the symmetric difference $|N(\mathbf{x})\Delta N(\mathbf{y})|$ to be $\Omega(n)$ not just for most pairs of vertices, but for most pairs of *pairs*. This is a stronger structural requirement than diversity, motivating the notion of richness which we will introduce in the next section.

Slightly abusing the notation, we write $U \cup \mathbf{Z}$ as a shorthand for $U \cup \bigcup_{\mathbf{z}\in\mathbf{Z}} \mathbf{z}$ whenever we are taking the union of a vertex set U and a possiple set of pairs $\mathbf{Z}$. Beyond showing $|\mathbf{X}| = \Omega(\sqrt{n})$, we also have to ensure $\mathbf{X}$ actually provides $\Theta(\sqrt{n})$ distinct degrees into $U \cup \mathbf{Z}$ for each choice of augmenting set $\mathbf{Z} \subseteq \mathbf{S} \cup \mathbf{T}$. Notice that the current construction of $\mathbf{X}$ only ensures distinct degrees $d_U(\mathbf{x})$ into U instead of $d_{U\cup Z}(\mathbf{x})$. If U were sampled once and directly constructed $\mathbf{S}, \mathbf{T}, \mathbf{X}$ based on U, then the $d_U(\mathbf{x})$ would be fixed numbers given the outcome of U. However, there is no guarantee that $d_{U\cup Z}(\mathbf{x}) = d_U(\mathbf{x}) + d_Z(\mathbf{x})$ would remain distinct, as the addition of $d_Z(\mathbf{x})$ could cancel out the uniqueness of $d_U(\mathbf{x})$. Kwan and Sudakov overcome this with a *double-exposure* technique. They perform two rounds of sampling. In the first round, they sampled the random set U_0 and construct $\mathbf{S}, \mathbf{T}, \mathbf{X}$ relative to U_0. In the second round, they sampled U from U_0 by independently keeping each vertex with probability 1/2. After the first sampling, the construction guarantees $|N_{U_0}(\mathbf{x})\Delta N_{U_0}(\mathbf{y})| = \Omega(n)$ for all $x, y \in \mathbf{X}$, and the augmenting sets $\mathbf{Z} \subseteq \mathbf{S}\cup\mathbf{T}$ are fully determined by U_0. In the second round of sampling, the difference $d_U(\mathbf{x}) - d_U(\mathbf{y})$ becomes a sum of $\Omega(n)$ independent Bernoulli terms, and Littlewood-Offord yields $\mathbb{P}[d_U(\mathbf{x}) - d_U(\mathbf{y}) = t] = O(1/\sqrt{n})$ for every integer t. In particular, since $t_{xy} = d_Z(\mathbf{y}) - d_Z(\mathbf{x})$ is a fixed integer for each $\mathbf{x}, \mathbf{y}$, we may bound for the collision probability

$$\mathbb{P}[d_{U\cup Z}(\mathbf{x}) = d_{U\cup Z}(\mathbf{y})] = \mathbb{P}[d_U(\mathbf{x}) - d_U(\mathbf{y}) = t_{xy}] = O(1/\sqrt{n}).$$

The same collision graph construction now finds us the desired set of U such that $d_{U\cup Z}(\mathbf{x})$ is distinct for $\mathbf{x} \in \mathbf{X}$.

3.1 Pair diversity and richness

We now introduce the structural conditions needed for the Kwan–Sudakov argument. Recall from Chapter 2 that a graph G is (ε, δ)-diverse if for each vertex x, at most n^δ vertices y have $|N(x)\Delta N(y)| < \varepsilon n$. In Chapter 2, diversity was the structural condition that guaranteed $|N(x)\Delta N(y)| = \Omega(n)$ for most pairs, which via anticoncentration gave the $O(1/\sqrt{n})$ collision probability for single vertices.

As discussed above, the proof of Theorem 3.1 runs the same anticoncentration mechanism but for pairs of vertices. This requires controlling the symmetric difference $N(\mathbf{x})\Delta N(\mathbf{y})$ for pairs $\mathbf{x} = \{x_1, x_2\}$, which is what *pair-diversity* provides.

Definition 3.1. An n-vertex graph G is $(\varepsilon, \delta, \alpha)_2$-*diverse* if for each pair of vertices $\mathbf{x} = \{x_1, x_2\}$ in G with $|N(x_1)\Delta N(x_2)| \geq \alpha n$, there are at most n^δ disjoint pairs of vertices $\mathbf{y} = \{y_1, y_2\}$ with $\mathbf{x} \cap \mathbf{y} = \emptyset$, such that $|N(\mathbf{x})\Delta N(\mathbf{y})| < \varepsilon n$.

We introduce a natural strengthening of diversity that also implies pair-diversity called *richness*.

Definition 3.2. An n-vertex graph G is (ε, δ)-*rich* if, for $W \subseteq V(G)$ with $|W| \geq \delta n$, we have $|N(v) \cap W| \geq \varepsilon |W|$ and $|\overline{N(v)} \cap W| \geq \varepsilon |W|$ for all but at most n^δ vertices $v \in V(G)$.

The key property of richness is that it controls the interaction of vertex neighbourhoods with arbitrary large vertex sets, rather than just with individual neighbourhoods. Diversity ensures that a large symmetric difference between two single-vertex neighbourhoods, whereas richness extends this property to pairs, as controlling the intersection of neighbourhoods with arbitrary large sets allows us to handle sets like $N(x_1) \cap \overline{N(x_2)}$ that arise when decomposing the pair-symmetric difference. The following lemma formalizes this idea.

Lemma 3.2. *Let an n-vertex graph G be (ε, δ)-rich, with $\delta \leq 1/2$. Then*

1. *G is $(\varepsilon/2, \delta)$-diverse,*
2. *G is $(\alpha\varepsilon/2, \delta, \alpha)_2$-diverse for any $\alpha \geq 2\delta$,*
3. *G has at most $n^{1+\delta}$ pairs of vertices $x_1, x_2 \in V(G)$ with $|N(x_1)\Delta\overline{N(x_2)}| < (\varepsilon/2)n$.*

Proof. We prove each statement in order.

1. Note that for each $x \in V(G)$, either $|N(x)| \geq n/2$ or $|\overline{N(x)}| \geq n/2$. If $|N(x)| \geq n/2$, then richness implies that
$$|\overline{N(y)} \cap N(x)| \geq \varepsilon |N(x)| \geq \varepsilon n/2$$
for at most n^δ vertices $y \in V(G)$. Similarly, if $|\overline{N(x)}| \geq n/2$, then
$$|N(y) \cap \overline{N(x)}| \geq \varepsilon |\overline{N(x)}| \geq \varepsilon n/2$$
for at most n^δ vertices $y \in V(G)$. But then in either case,
$$|N(x)\Delta N(y)| = |\overline{N(y)} \cap N(x)| + |N(y) \cap \overline{N(x)}| \geq \varepsilon n/2.$$

2. Suppose not. Then there exists a pair $\mathbf{x} = \{x_1, x_2\}$ with $|N(x_1)\Delta\overline{N(x_2)}| \geq \alpha n$ and a set Y of n^δ pairs such that $|N(\mathbf{x})\Delta N(\mathbf{y})| < \alpha\varepsilon n/2$ for all $\mathbf{y} \in Y$. Since
$$|N(x_1)\Delta\overline{N(x_2)}| = |N(x_1) \cap N(x_2)| + |\overline{N(x_1)} \cap \overline{N(x_2)}| \geq \alpha n,$$

we have $|N(x_1) \cap N(x_2)| \geq (\alpha/2)n \geq \delta n$ or $|\overline{N(x_1)} \cap \overline{N(x_2)}| \geq (\alpha/2)n \geq \delta n$. Assume without loss of generality that the former case holds. Then for each vertex y in any $\mathbf{y} \in Y$,

$$\left|\overline{N(y)} \cap (N(x_1) \cap N(x_2))\right| \leq |N(\mathbf{x}) \Delta N(\mathbf{y})| < \alpha\varepsilon n/2 \leq \varepsilon |N(x_1) \cap N(x_2)|.$$

But then the set of all such y now contradicts the (ε, δ)-rich condition.

3. We show that for each choice of $x_1 \in V(G)$, there are at most n^δ vertices $x_2 \in V(G)$ with $|N(x_1) \Delta \overline{N(x_2)}| < (\varepsilon/2)n$. Since either $|N(x_1)| \geq n/2$ or $|\overline{N(x_1)}| \geq n/2$, we assume without loss of generality that $|N(x_1)| \geq n/2$. By (ε, δ)-richness, there are at most n^δ vertices x_2 with $|N(x_2) \cap N(x_1)| < \varepsilon n/2$. For all other choices of x_2,

$$|N(x_1) \Delta \overline{N(x_2)}| \geq |N(x_1) \cap N(x_2)| \geq (\varepsilon/2)n.$$

The result now follows from the fact that there are only n choices of x_1.

□

Remark: Note that part 3 of the lemma is the reversed notion of diversity. Since $|N(x_1) \Delta N(x_2)| + |N(x_1) \Delta \overline{N(x_2)}| = n - 2$ for any two vertices x_1, x_2, combining part 1 and 3 of the lemma yields a two-sided bound

$$(\varepsilon/2)n \leq |N(x_1) \Delta N(x_2)| \leq (1 - \varepsilon/2)n,$$

for most pairs of vertices x_1, x_2. In particular, $d(x_1) + d(x_2) = \Theta(n)$.

We also show the parallel result of Lemma 2.3 that every C-Ramsey graph contains an (ε, δ)-rich subgraph of linear size, following the same line of argument.

Lemma 3.3. *For any $C > 0$ and $\delta \in (0, 1)$, there exists $\varepsilon = \varepsilon(C, \delta) > 0$ and $c = c(C, \delta) > 0$ such that every C-Ramsey graph G on n-vertex contains an (ε, δ)-rich subgraph on $\Omega(n)$ vertices.*

Proof. Let $K = K(C)$ be a large constant to be determined. Let $c \leq (\delta/4)^K$, and $\varepsilon = 1/K$. Suppose for contradiction that any vertex subset of size at least cn fails to induce an (ε, δ)-rich subgraph. We will inductively construct a sequence of disjoint vertex sets $S_1, \ldots, S_K$ of size $(cn)^\delta$, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 4\varepsilon$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 4\varepsilon$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G_0 \subset G_1 \subset \cdots \subset G_K$, such that $|V(G_i)| \geq (\delta/4)|V(G_{i-1})|$ and $S_i \subseteq V(G_{i-1})$ for all i.

We first show that this construction suffices. By working with $\overline{G}$ instead of G if necessary, we may assume that $r \geq K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^{r} S_{i_k}$. Since each S_{i_k} has size $(cn)^\delta$ and $\varepsilon = 1/K$,

$$d(S) \leq \frac{\binom{(cn)^\delta}{2}}{\binom{r(cn)^\delta}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \leq \frac{1}{r} + 4\varepsilon \leq \frac{6}{K}.$$

Note that $|S| \geq (cn)^\delta$, and so Theorem 1.3 yields a constant $\gamma > 0$ and a homogeneous set A in S of size

$$|A| \geq \frac{\gamma K}{6\log(K/6)} \log|S| \geq \frac{\gamma\delta K}{6\log(K/6)} \log cn = \frac{\gamma\delta K}{6\log(K/6)} \log n + o(\log n),$$

as $n \to \infty$. But then $\frac{\gamma\delta K}{6\log(K/6)} > C$ for sufficiently large K, and so $|A| > C\log n$, contradiction.

We now construct the sets S_i and G_i inductively. Suppose $G_1, \ldots, G_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. Since $|V(G_{i-1})| \geq (\delta/4)^{i-1}n \geq cn$ and G_{i-1} is not (ε, δ)-rich by assumption, there exist $W_i, Y_i \subseteq V(G_{i-1})$ with $|W_i| \geq \delta|V(G_{i-1})|$ and $|Y_i| \geq (cn)^\delta$, such that for each $y \in Y_i$ we have $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ or $|\overline{N_{G_{i-1}}(y)} \cap W_i| < \varepsilon|W_i|$. Assume without loss of generality that $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ for half the vertices $y \in Y_i$, and let S_i be the corresponding subset of Y_i. Put $T_i = W_i \setminus S_i$ and note that $|T_i| \geq |W_i|/2$. Partition T_i into $V_i \sqcup Z_i$ where $V_i = \{v \in T_i : d(v, S_i) \leq 4\varepsilon\}$, and let $G[V_i]$. We need to show that $|V_i| \geq (\delta/4)|V(G_{i-1})|$. Observe that for all $y \in S_i$,

$$d(y, T_i) = \frac{d_{T_i}(y)}{|T_i|} < \frac{\varepsilon|W_i|}{|W_i|/2} = 2\varepsilon.$$

But then

$$4\varepsilon|Z_i| < \sum_{v \in Z_i} d(v, S_i) \leq \frac{e(T_i, S_i)}{|S_i|} = \frac{|T_i|}{|S_i|} \sum_{y \in S_i} d(y, T_i) \leq 2\varepsilon|T_i|,$$

so $|Z_i| < |T_i|/2$. It now follows that $|V_i| \geq |T_i|/2 \geq |W_i|/4 \geq (\delta/4)|V(G_{i-1})|$. Since $d(S_i, U) < 4\varepsilon$ for all $U \subseteq V_i$, condition (i) holds for S_i. In the case that $|\overline{N_{G_{i-1}}(x_1)} \cap W_i| < \varepsilon|W_i|$, set $V_i = \{v \in T_i : d(v, S_i) > 1 - 4\varepsilon\}$, which yields $|V_i| \geq (\delta/4)|G_{i-1}|$ and S_i satisfying condition (ii). This completes the induction. □

3.2 The main lemma and the construction

As outlined above, we will find many induced subgraphs $G[U]$ whose edge counts are well-separated from each other, and show that each can be augmented in $\Omega(n^{3/2})$ ways. Theorem 3.1 will be an immediate consequence of the following lemma, which we will dedicate the rest of this chapter to prove.

Lemma 3.4. *Let $C > 0$. Then there exists $c = c(C) > 0$, such that for any C-Ramsey graph G on n vertices and any m satisfying $cn^2 \le m \le 2cn^2$, there are disjoint subsets $U, W \subseteq V(G)$ with $|e(U) - m| = O(n^{3/2})$ and $|W| = O(\sqrt{n})$ such that*

$$|\{e(U \cup Z) : Z \subseteq W\}| = \Omega(n^{3/2}).$$

We first show how this implies Theorem 3.1.

Proof of Theorem 3.1. Note that for any U, W given by Lemma 3.4,

$$|e(U \cup W) - e(U)| = O(n^{3/2}),$$

as $|W| = O(\sqrt{n})$ and each $w \in W$ has at most n neighbours in $U \cup W$. But then the $\Omega(n^{3/2})$ values of $e(U \cup Z)$ for $Z \subseteq W$ are contained in an interval of length $O(n^{3/2})$ centered at $e(U)$. Partition the interval between cn^2 and $2cn^2$ into $\frac{c}{k}\sqrt{n}$ subintervals of length $kn^{3/2}$, for some large enough constant k to be determined. Now apply Lemma 3.4 to a suitable choice of m in each subinterval so that the obtained values of $e(U \cup Z)$ for $Z \subseteq W$ lie in the same subinterval as m, which is allowed for sufficiently large k. Since we applied Lemma 3.4 to $\Omega(\sqrt{n})$ disjoint subintervals, we obtain $\Omega(n^2)$ distinct sizes of induced subgraphs. □

3.3 The construction

We now prove the key construction that produces the sets $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}$ with the properties needed for Lemma 3.4. The following lemma is due to Kwan and Sudakov [11, Lemma 7], and its proof combines the structural tools developed in Section 3.1 with the anticoncentration mechanism from Section 1.2.

Lemma 3.5. *For any $C > 0$ there exists $c = c(C) > 0$ such that the following holds. For any C-Ramsey graph G on n vertices and any m satisfying $cn^2 \le m \le 2cn^2$, there is a vertex set $U_0 \subseteq V(G)$ with $|e(U_0) - 4m| = O(n^{3/2})$, a constant $c' = c'(C) > 0$, and sets $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$ of size $c'\sqrt{n}$ such that*

1. *either $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X} \subseteq V(G)$ are disjoint sets of vertices, or $\mathbf{S}, \mathbf{T}, \mathbf{X} \subseteq \binom{V(G)}{2}$ are sets of disjoint pairs of vertices with no vertex appearing in more than one of $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}$,*
2. *there is $d = \Theta(n)$ such that $d_{U_0}(\mathbf{x}) = d + O(\sqrt{n})$ for each $\mathbf{x} \in \mathbf{S} \cup \mathbf{T} \cup \mathbf{X}$,*
3. *$\min_{\mathbf{x} \in \mathbf{T}} d_{U_0}(\mathbf{x}) - \max_{\mathbf{x} \in \mathbf{S}} d_{U_0}(\mathbf{x}) \ge 8c'\sqrt{n}$,*
4. *for each $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{X}}{2}$, we have $|N_{U_0}(\mathbf{x}) \Delta N_{U_0}(\mathbf{y})| = \Omega(n)$.*

The four properties correspond to the four roles played by $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$ in the counting argument of Section 3.4. Property 2 ensures that each element contributes $\Theta(n)$ edges to any subgraph containing U_0, so that adding or removing a single element shifts the edge count by $\Theta(n)$. Property 3 guarantees that swapping an element of $\mathbf{S}$ for an element of $\mathbf{T}$ shifts the edge count by $\Theta(\sqrt{n})$. Property 4 is the anticoncentration

condition on $\mathbf{X}$, which ensures that elements of $\mathbf{X}$ are likely to have many distinct degrees into $U \cup Z$ after the double-exposure step.

The proof will start with reducing to a rich subgraph of linear size using Lemma 3.3 and using the pigeonhole principle to obtain a large candidate set $\mathbf{W}_0$ of vertices or pairs with similar degrees, satisfying Property 1 and Property 2. We will then sample U_0 at random and verify that all desired properties hold simultaneously with positive probability.

Proof of Lemma 3.5. Let $\varepsilon = \varepsilon(C)$ be given by Lemma 3.3 and set $\delta = \varepsilon/4$. By Lemma 3.3, G contains an (ε, δ)-rich induced subgraph $G[V']$ on $\Omega(n)$ vertices. Since $\log |V'| \geq \frac{1}{2} \log n$, the graph $G[V']$ is $2C$-Ramsey, so by adjusting constants it suffices to work inside $G[V']$. In particular, we may assume G itself is (ε, δ)-rich.

By Lemma 3.2 with $\alpha = \varepsilon/2$, this richness assumption furnishes the following three properties:

(a) G is $(\varepsilon/2, \delta)$-diverse,

(b) G is $(\varepsilon^2/4, \delta, \varepsilon/2)_2$-diverse,

(c) there are at most $n^{1+\delta}$ pairs $\{x_1, x_2\}$ with $|N(x_1)\Delta\overline{N(x_2)}| < (\varepsilon/2)n$.

Each of the $\binom{n}{2} = \Theta(n^2)$ vertex pairs $\{x_1, x_2\}$ has a pair-degree $d(\{x_1, x_2\}) = d(x_1) + d(x_2)$ taking values from 0 to $2n$, and so by the pigeonhole principle there exists a value d^* and a collection H of $\Omega(n^{3/2})$ pairs with $d(\mathbf{x}) = d^* + O(\sqrt{n})$ for all $\mathbf{x} \in H$. By the remark after Lemma 3.2, $d^* = \Theta(n)$. Interpret H as a graph on $V(G)$ whose edges are the $\Omega(n^{3/2})$ pairs. By (c), we may delete $O(n^{1+\delta}) = o(n^{3/2})$ edges in H so that all remaining pairs have $|N(x_1)\Delta\overline{N(x_2)}| \geq (\varepsilon/2)n$. Suppose there exists vertex v in H with degree $\Omega(n^{3/4})$. Take $\mathbf{W}_0 = N_H(v)$ and set $d = d^* - d(v)$. Then each $w \in \mathbf{W}_0$ is a single vertex with degree $d(w) = d + O(\sqrt{n})$. Now suppose H has maximum degree $\Delta = o(n^{3/4})$. By iteratively picking an edge and removing at most 2Δ edges adjacent to it, we obtain a matching of size $\Omega(n^{3/2})/o(n^{3/4}) = \Omega(n^{3/4})$. Take $\mathbf{W}_0$ to be the pairs in this matching and set $d = d^*$. In either case, the subsets of $\mathbf{W}_0$ satisfy Property 1 and Property 2.

Next, we extract a subset $\mathbf{A} \subseteq \mathbf{W}_0$ of size $\Omega(\sqrt{n})$ with pairwise large symmetric differences. Define graph F on $\mathbf{L}$ by joining $\mathbf{x}$ and $\mathbf{y}$ whenever $|N(\mathbf{x})\Delta N(\mathbf{y})| < (\varepsilon^2/4)n$. By diversity, the graph F has $|\mathbf{W}_0| = \Omega(n^{3/4})$ vertices and maximum degree at most $n^\delta < n^{1/4}$. Thus by Turán's theorem, F has an independent set $\mathbf{A} \subseteq \mathbf{W}_0$ with size $\Omega(|\mathbf{W}_0|/n^{1/4}) = \Omega(\sqrt{n})$. That is, for $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$, we have $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$.

By Corollary 1.4 and the Ramsey property, $e(G) \geq 800cn^2$ for some constant $c = c(C) > 0$. For $cn^2 \leq m \leq 2cn^2$, set $p = \sqrt{4m/e(G)}$, so that $p = \Omega(1)$ and $p \leq 0.1$. Let U_0 be a random subset of $V(G)$ formed by including each vertex independently with probability p. We make the following claim:

Claim 3.6. *The following five events each hold with probability greater than* $4/5$*, and therefore all hold simultaneously with positive probability by the union bound.*

(i) $|e(U_0) - 4m| = O(n^{3/2})$*,*

(ii) There is a set $\mathbf{Q} \subseteq \mathbf{A}$ *involving no vertices of* U_0*, with* $|\mathbf{Q}| \geq (2/3)|\mathbf{A}|$*,*

(iii) There is a set $\mathbf{R} \subseteq \mathbf{A}$ *with* $|\mathbf{R}| \geq (2/3)|\mathbf{A}|$ *and* $d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})$ *for each* $\mathbf{x} \in \mathbf{R}$*,*

(iv) $|N_{U_0}(\mathbf{x}) \Delta N_{U_0}(\mathbf{y})| = \Omega(n)$ *for each* $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$*,*

(v) The equality $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$ *holds for* $O(\sqrt{n})$ *pairs* $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$*.*

Proof of Claim. We prove each part in order.

(i) Note that $\mathbb{E}[e(U_0)] = 4m$ and there are $O(n^3)$ pairs of edges in G whose presense are dependent in $G[U_0]$, as this only occur if two edges share a vertex. For $e \in E(G)$, let A_e be the event that e is present in $G[U_0]$. Since

$$\mathrm{Var}(e(U_0)) \leq \mathbb{E}[e(U_0)] + \sum_{e \sim e'} \mathbb{P}(A_e \cap A_{e'}) = O(n^3),$$

it follows from Chebyshev's inequality that

$$\mathbb{P}[|e(U_0) - 4m| = \omega(n^{3/2})] \leq \frac{O(n^3)}{\omega(n^3)} \to 0,$$

as $n \to \infty$.

(ii) Let $\mathbf{Q}$ be the subset of $\mathbf{A}$ involving no vertices of U_0. Then $|\mathbf{Q}|$ follows a binomial distribution with $|\mathbf{A}|$ trials, it has mean at least $(1-p)^2|\mathbf{A}| \geq 0.8|\mathbf{A}|$ and variance $O(|\mathbf{A}|)$. It follows from Chebyshev's inequality that

$$\mathbb{P}\left[|\mathbf{Q}| < \frac{2}{3}|\mathbf{A}|\right] \leq \mathbb{P}\left[|\mathbf{Q}| - \mathbb{E}[|\mathbf{Q}|] \geq 0.1|\mathbf{A}|\right] \leq \frac{O(|\mathbf{A}|)}{0.01|\mathbf{A}|^2} = O(1/|\mathbf{A}|) \to 0,$$

as $n \to \infty$.

(iii) For $\mathbf{x} \in |\mathbf{A}|$, the degree $d_{U_0}(\mathbf{x})$ has mean $pd + O(\sqrt{n})$ and variance $O(n)$. By Chebyshev's inequality,

$$\mathbb{P}[d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})] \leq 1 - \mathbb{P}[|d_{U_0}(\mathbf{x}) - \mathbb{E}[d_{U_0}(\mathbf{x})]| = \omega(\sqrt{n})] \leq 1 - \frac{O(n)}{\omega(n)} \to 1.$$

In particular, $d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})$ with probability 0.99 when n is sufficiently large. Let $\mathbf{R} \subseteq \mathbf{A}$ be the set of $\mathbf{x}$ that satisfies this bound. Since $\mathbb{E}[|\mathbf{A} \setminus \mathbf{R}|] = 0.01|\mathbf{A}|$. It now follows from Markov's inequality that

$$\mathbb{P}\left[|\mathbf{R}| \geq \frac{2}{3}|\mathbf{A}|\right] = 1 - \mathbb{P}\left[|\mathbf{A} \setminus \mathbf{R}| \geq \frac{1}{3}|\mathbf{A}|\right] \geq 1 - \frac{\mathbb{E}[|\mathbf{A} \setminus \mathbf{R}|]}{\frac{1}{3}|\mathbf{A}|} \geq 0.8.$$

(iv) Recall that $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$ for $\{\mathbf{x},\mathbf{y}\} \in \binom{\mathbf{A}}{2}$. Since $|N_{U_0}(\mathbf{x})\Delta N_{U_0}(\mathbf{y})| = |N(\mathbf{x})\Delta N(\mathbf{y}) \cap U_0|$, it has a binomial distribution with parameters $N_{U_0}(\mathbf{x})$ and p. By the Chernoff bound,

$$\mathbb{P}\left[|N_{U_0}(\mathbf{x})\Delta N_{U_0}(\mathbf{y})| < \frac{p}{2}|N(\mathbf{x})\Delta N(\mathbf{y})|\right] = e^{-\Omega(n)} = o(n)$$

The result now follows from the union bound.

(v) For $\{\mathbf{x},\mathbf{y}\} \in \binom{\mathbf{A}}{2}$, note that the random variable $d_{U_0}(\mathbf{x}) - d_{U_0}(\mathbf{y})$ is of $(|N((x))\Delta N((y)), p|)$-Littlewood-Offord type. Since $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$, we have $\mathbb{P}[d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})] = O(1/\sqrt{n})$, and so the expected number of paris $\{\mathbf{x},\mathbf{y}\} \in \binom{\mathbf{A}}{2}$ satisfying $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$ is thus $O(\sqrt{n})$. The result now follows from Markov's inequality.

□

Fix an outcome of U_0 for which all five events hold. Let $\mathbf{A}' = \mathbf{Q} \cap \mathbf{R}$, and note that $|\mathbf{A}'| \geq |\mathbf{A}|/3 = \Omega(\sqrt{n})$. It remains to obtain $\mathbf{S}$, $\mathbf{T}$, and $\mathbf{X}$ from $\mathbf{A}'$ that satisfy Properties 2–4. Consider the graph on $\mathbf{A}'$ where $\mathbf{x}$ and $\mathbf{y}$ are adjacent if and only if $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$. By Turán Theorem (Proposition 1.5), there is an independent set $\mathbf{W} \subseteq \mathbf{A}'$ of size $\Omega(\sqrt{n})$, and note that $\mathbf{W}$ has distinct degrees into U_0. Partition $\mathbf{W}$ into three equal parts by the degree $d_{U_0}(\mathbf{x})$ into U_0, and let $\mathbf{T}$ be the top third, $\mathbf{S}$ be the bottom third, and $\mathbf{X}$ be the middle third. Since $\mathbf{W}$ has distinct degrees and $|\mathbf{X}| = \Omega(\sqrt{n})$, we may find constant c' that satisfies Property 3. Since $\mathbf{X} \subseteq \mathbf{A}$, Property 4 holds. Trimming the sizes of $\mathbf{S}, \mathbf{T}, \mathbf{X}$ to $c'\sqrt{n}$ completes the proof. □

Remark: Claim **??** is where the anticoncentration principle from Section 1.2 enters the proof. For each pair $\mathbf{x}, \mathbf{y} \in \mathbf{A}$ with $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$, the difference $d_{U_0}(\mathbf{x}) - d_{U_0}(\mathbf{y})$ is a sum of $\Omega(n)$ independent bounded random variables (since U_0 is formed by independent Bernoulli sampling). By the Erdős–Littlewood–Offord theorem, the probability that this sum equals zero is $O(1/\sqrt{n})$. This is the Bernoulli-sampling case of the anticoncentration mechanism discussed in Section 1.2, in contrast to the hypergeometric case used in Lemma 2.2.

3.4 The double-exposure argument

Given $\mathbf{S}, \mathbf{T}, c'$ from Lemma 3.5, fix an ordering on $\mathbf{S}$ and $\mathbf{T}$ and define

$$\mathcal{P} := \{(k,i) : c'\sqrt{n} \leq k \leq 2c'\sqrt{n} \text{ and } 0 \leq i \leq c'\sqrt{n}\}.$$

For $(k,i) \in \mathcal{P}$, define $\mathbf{Z}_{k,i} \subseteq \mathbf{S} \sqcup \mathbf{T}$ by taking the first $k-i$ elements from $\mathbf{S}$ and the first i elements from $\mathbf{T}$. By Property 2 and 3 of Lemma 3.5, the values of $e(U_0, \mathbf{Z}_{k,i}) := \sum_{\mathbf{z}\in\mathbf{Z}_{k,i}} d_{U_0}(\mathbf{z})$ are evenly spaced by $\Theta(\sqrt{n})$ as (k,i) vary lexicographically, which already forms $\Omega(n)$ distinct values of $e(U_0 \cup \mathbf{Z}_{k,i})$. What remains is to fill the $\Theta(\sqrt{n})$ gaps between each $e(U_0 \cup \mathbf{Z}_{k,i})$ with $\mathbf{X}$ via the *double-exposure* technique while

maintaining the nice gap struture of the subsets $\mathbf{Z}_{k,i}$. More specifically, we sample U from U_0 such that the values $e(U \cup \mathbf{Z}_{k,i} \cup \{\mathbf{x}\})$ are distinct as $\mathbf{Z}_{k,i} \subseteq \mathbf{S} \sqcup \mathbf{T}$ and $\mathbf{x} \in \mathbf{X}$ vary. Define $U_{k,i} \coloneqq U \cup \mathbf{Z}_{k,i}$ and $e_{k,i} \coloneqq e(U_{k,i}) - e(U) = e(\mathbf{Z}_{k,0}) + e(\mathbf{Z}_{k,0}, U)$. There are four things we need to ensure:

1. *Degree distinctness in* $\mathbf{X}$. For most pairs (k, i), the elements of $\mathbf{X}$ should have $\Omega(\sqrt{n})$ pairwise distinct degrees into $U_{k,i}$, all lying in an interval of length $O(\sqrt{n})$. This is the payoff of double-exposure.

2. *Concentration of the edge count.* The value $e_{k,0}$ should concentrate around its expectation within $O(n)$, so that the $\Theta(n)$ spacing between $e_{k,0}$ and $e_{k+1,0}$ are perserved.

3. *Regular medium-scale spacing.* For each k, the values $e_{k,0}, e_{k,1}, \ldots, e_{k,c'\sqrt{n}}$ should span a range of $\Omega(n)$ and increase roughly uniformly, so that $\Theta(\sqrt{n})$ of them are pairwise separated by $\Omega(\sqrt{n})$.

Write $\Delta_{k,i} \coloneqq e_{k,i} - e_{k,i-1}$ and we have the following lemma, whose first two properties correspond to the first two points above, and the last two properties combined ensures regular medium-scale spacing.

Lemma 3.7. *Let G be a C-Ramsey graph, let $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$, U_0, c', d be obtained from Lemma 3.5. Let U be a random subset of U_0 where each vertex is present with probability $1/2$, and let $U_{k,i}, e_{k,i}, \Delta_{k,i}$ be defined as above. Then there exist positive constants $M, \beta, Q > 0$ depending on C, such that $Q \geq 3\beta$ and for each $c'\sqrt{n} \leq k \leq 2c'\sqrt{n}$ the following hold:*

1. *With probability at least 0.99, there is a set $\mathcal{I}_k$ of $(1 - \beta/(2M)) \cdot c'\sqrt{n}$ values with the following property: for each $i \in \mathcal{I}_k$, there exists $\mathbf{X}_{k,i} \subseteq \mathbf{X}$ of size $\Omega(\sqrt{n})$, such that the values $d_{U_{k,i}}(\mathbf{x})$ for $\mathbf{x} \in \mathbf{X}_{k,i}$ are distinct and all lie in the interval $[d/2 - Q\sqrt{n},\ d/2 + Q\sqrt{n}]$.*

2. *With probability at least 0.99,*
$$|e_{k,0} - \mathbb{E}[e_{k,0}]| \leq Qn.$$

3. *With probability at least $1/2$,*
$$e_{k,c'\sqrt{n}} - e_{k,0} \geq 3\beta n.$$

4. *With probability at least 0.99,*
$$\sum_{i\,:\,|\Delta_{k,i}| \geq M\sqrt{n}} |\Delta_{k,i}| \leq \beta n.$$
That is, the unusually large increments contribute little to the total range $e_{k,c'\sqrt{n}} - e_{k,0}$.

Proof. We prove each property in the order of 3, 4, 1, 2, so that the constants β, M, Q may be determined in their dependency order.

3. Note that $|e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})| \le 2c'\sqrt{n}$, as $|\mathbf{Z}_{k,i}| = k \le 2c'\sqrt{n}$. This and the construction of U yields

$$\begin{aligned}\mathbb{E}[e_{k,i} - e_{k,i-1}] &= \mathbb{E}[e(U, \mathbf{Z}_{k,i}) - e(U, \mathbf{Z}_{k,i-1})] + [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})] \\ &\ge \frac{1}{2}[e(U_0, \mathbf{Z}_{k,i}) - e(U_0, \mathbf{Z}_{k,i-1})] - 2c'\sqrt{n}.\end{aligned}$$

It follows form Property 3 of Lemma 3.5 that $e(U_0, \mathbf{Z}_{k,i}) - e(U_0, \mathbf{Z}_{k,i-1}) \ge 8c'\sqrt{n}$, and so

$$\mathbb{E}[e_{k,i} - e_{k,i-1}] \ge 2c'\sqrt{n},$$

for all i. Put $\beta = 2(c')^2/3$ so the difference $e_{k,c'\sqrt{n}} - e_{k,0}$ has expectation at least $3\beta n$. But then since $e(U, \mathbf{Z}_{k,i}) = \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot d_{\mathbf{Z}_{k,i}}(u)$,

$$\begin{aligned}e_{k,c'\sqrt{n}} - e_{k,0} &= [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,0})] + [e(U, \mathbf{Z}_{k,i}) - e(U, \mathbf{Z}_{k,0})] \\ &= [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,0})] + \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot (d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,0}}(u))\end{aligned}$$

is of Littlewood-Offord type with probability 1/2, which is symmetrically distributed around its expectation. The result now follows.

4. Note that

$$\Delta_{k,i} = [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})] + \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot (d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,i-1}}(u))$$

is of Littlewood-Offord type and measures the difference of edge count after swapping exactly one vertex from $\mathbf{S}$ to $\mathbf{T}$, so by Lemma 3.5 the expectation of $\Delta_{k,i}$ is $O(\sqrt{n})$ and each $u \in U_0$ contributes at most $|d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,i-1}}(u)| \le 2$ to $\Delta_{k,i}$. Thus, the Hoeffding's inequality yields

$$\mathbb{P}[|\Delta_{k,i}| \ge t] \le 2\exp\left(\frac{2(t - \mathbb{E}[|\Delta_{k,i}|])^2}{4|U_0|}\right) = \exp(-\Omega(t^2/n)).$$

Using the identity $\mathbb{E}[X] = \sum_{t \ge 1} \mathbb{P}(X \ge t)$ for any nonnegative integer random variable X, we get

$$\begin{aligned}\mathbb{E}\left[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}| \ge M\sqrt{n}}\right] &= \sum_{t=1}^{\infty} \mathbb{P}\left[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}| \ge M\sqrt{n}} \ge t\right] \\ &= M\sqrt{n} \cdot \mathbb{P}\left[|\Delta_{k,i} \ge M\sqrt{n}|\right] + \sum_{t=M\sqrt{n}}^{\infty} \mathbb{P}\left[|\Delta_{k,i}| \ge t\right] \\ &= M\sqrt{n} \cdot \exp(-\Omega(M^2)) + \sum_{t=M\sqrt{n}}^{\infty} \exp(-\Omega(t^2/n)) \\ &= \sqrt{n} \cdot \exp(-\Omega(M^2)),\end{aligned}$$

uniformly over M. It now follows from Markov's inequality that, for large enough M,

$$\mathbb{P}\left[\sum_{i\,:\,|\Delta_{k,i}|\geq M\sqrt{n}} |\Delta_{k,i}| > \beta n\right] \leq \frac{\sum_{i=1}^{|\mathbf{S}|}\mathbb{E}\left[|\Delta_{k,i}|\cdot \mathbb{1}_{|\Delta_{k,i}|\geq M\sqrt{n}}\right]}{\beta n} \leq \frac{e^{-\Omega(M^2)}}{\beta} < 0.99.$$

1. For each $\mathbf{x},\mathbf{y}\in\binom{\mathbf{X}}{2}$, the difference

$$d_U(\mathbf{x}) - d_U(\mathbf{y}) = \sum_{u\in N_U(\mathbf{x})}\mathbb{1}_{u\in U} - \sum_{u\in N_U(\mathbf{y})}\mathbb{1}_{u\in U} = \sum_{u\in N_U(\mathbf{x})\Delta N_U(\mathbf{y})} a_u\mathbb{1}_{u\in U}$$

with $a_u\in\{1,-1\}$. Thus for each k,i, the random variable $d_{U_{k,i}}(\mathbf{x}) - d_{U_{k,i}}(\mathbf{y})$ is of $(|N_U(\mathbf{x})\Delta N_U(\mathbf{y})|, 1/2)$-Littlewood-Offord type, as it is simply translation of $d_U(\mathbf{x}) - d_U(\mathbf{y})$ by some deterministic value. It follows from Property 4 of Lemma 3.5 and Littlewood-Offord theorem (Proposition 1.8) that

$$\mathbb{P}[d_{U_{k,i}}(\mathbf{x}) = d_{U_{k,i}}(\mathbf{y})] = O(1/\sqrt{n}).$$

Let $H_{k,i}$ be the graph on $\mathbf{X}$ with edges $\{\mathbf{x},\mathbf{y}\}\in\binom{\mathbf{X}}{2}$ such that $d_{U_{k,i}}(\mathbf{x}) = d_{U_{k,i}}(\mathbf{y})$, so we have $\mathbb{E}[e(H_{k,i})] = |\mathbf{X}|^2\cdot O(1/\sqrt{n}) = O(\sqrt{n})$. By Markov's inequality, for large enough n,

$$\mathbb{P}[e(H_{k,i}) = O(\sqrt{n})] \geq 1 - \frac{\mathbb{E}[e(H_{k,i})]}{\omega(\sqrt{n})} \geq 1 - \frac{\beta}{400M},$$

in which case Turán's theorem (Proposition 1.5) yields an independent set $\mathbf{Y}_{k,i}$ of size $n/(2O(\sqrt{n})+\sqrt{n}) = 2\gamma\sqrt{n}$, for some constant $\gamma = \gamma(\beta, M) > 0$. In particular, $d_{U_{k,i}}(\mathbf{x})$ is distinct for $\mathbf{x}\in\mathbf{Y}_{k,i}$. The expected proportion of values of i for which this fails to occur is $\beta/400M$, and by Markov's inequality again, it fails only for $\beta/2M$ proportion of values of i with probability at least $1-(\beta/400M)/(\beta/2M) = 0.995$, in which case put $\mathcal{I}_k$ as the set of i that succeeds.

By Property 2 of Lemma 3.5 we have $\mathbb{E}[d_U(\mathbf{x})] = d/2 + O(\sqrt{n})$ for each $\mathbf{x}\in\mathbf{X}$. Since $d_U(\mathbf{x})$ follows a binomial distribution with parameters $O(n)$ and $1/2$, we also have $\mathrm{Var}(d_U(\mathbf{x})) = O(n)$. Note that by triangle inequality,

$$|d_U(\mathbf{x}) - d/2| \leq |d_U(\mathbf{x}) - \mathbb{E}[d_U(\mathbf{x})]| + O(\sqrt{n}),$$

and so for sufficiently large Q, Chebyshev's inequality yields

$$\mathbb{P}\left[|d_U(\mathbf{x}) - d/2| > Q\sqrt{n}\right] \leq \frac{\mathrm{Var}(d_U(\mathbf{x}))}{Q^2 n} = \frac{O(1)}{Q^2} < \frac{\gamma}{200}.$$

Let $\mathbf{Y}$ be the set of all $\mathbf{x}\in\mathbf{X}$ such that $|d_U(\mathbf{x}) - d/2| \leq Q\sqrt{n}$. Then each $\mathbf{x}\in\mathbf{X}$ is not present in $\mathbf{Y}$ with probability less than $\gamma/200$. By Markov's inequality,

$$\mathbb{P}\left[|\mathbf{Y}| \geq (1-\gamma)|\mathbf{X}|\right] = 1 - \mathbb{P}\left[|\mathbf{X}\setminus\mathbf{Y}| \geq \gamma|\mathbf{X}|\right] \geq 1 - \frac{\gamma|\mathbf{X}|}{200\gamma|\mathbf{X}|} = 0.995.$$

In other words, with probability at least 0.995, there exists large enough constant Q and a set $\mathbf{Y} \subseteq \mathbf{X}$ of size at least $(1-\gamma)|\mathbf{X}|$ whose elements all lie in the interval $[d/2 - Q\sqrt{n},\ d/2 + Q\sqrt{n}]$.

By the union bound, both above events occur with probability at least 0.99, in which case we take $\mathbf{X}_{k,i} = \mathbf{Y}_{k,i} \cap \mathbf{Y}$ for $i \in \mathcal{I}_k$. Since $|\mathbf{Y}_{k,i}| \geq 2\gamma\sqrt{n}$ and $|\mathbf{Y}| \geq (1-\gamma)|\mathbf{X}|$, we have $|\mathbf{X}_{k,i}| \geq \gamma\sqrt{n}$. This proves Property 1.

2. Since
$$e_{k,i} = e(\mathbf{Z}_{k,i}) + e(U, \mathbf{Z}_{k,i}) = e(\mathbf{Z}_{k,i}) + \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot d_{\mathbf{Z}_{k,i}}(u),$$
the random variable $e_{k,i}$ is of $(O(n), 1/2)$-Littlewood-Offord type with coefficients $d_{\mathbf{Z}_{k,i}}(u) = O(\sqrt{n})$. It follows that $\mathrm{Var}(e_{k,0}) = O(n^2)$ and by Chebyshev's inequality,
$$\mathbb{P}\left[|e_{k,0} - \mathbb{E}[e_{k,0}]| \leq Qn\right] \geq 1 - \frac{\mathrm{Var}(e_{k,0})}{Q^2 n^2} = 0.99,$$
for large enough Q. We end this proof by noting that enlarging Q does not fail Property 1.

□

3.5 Proof of the main lemma

We now assemble Lemma 3.5 and Lemma 3.7 to prove the main lemma.

Apply Lemma 3.5 to obtain $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}, c', d$, and define $\mathcal{P}, U, \mathbf{Z}_{k,i}, U_{k,i}$ as in Section 3.4. Note that U is the only source of randomness. We prove the main lemma with W being the set of all vertices in $\mathbf{S}, \mathbf{T}, \mathbf{X}$.

Recall that k ranges over the interval between $c'\sqrt{n}$ and $2c'\sqrt{n}$. For each k, union bound shows that all four properties of Lemma 3.7 hold with probability at least 0.4. Let $\mathcal{K}$ be the set of all such k. Then $\mathbb{E}\left[c'\sqrt{n} - |\mathcal{K}|\right] \leq 0.6\, c'\sqrt{n}$, and Markov's inequality yields

$$\mathbb{P}\left[|\mathcal{K}| \geq 0.2\, c'\sqrt{n}\right] = 1 - \mathbb{P}\left[c'\sqrt{n} - |\mathcal{K}| \geq 0.8\, c'\sqrt{n}\right] \geq 1 - \frac{\mathbb{E}\left[c'\sqrt{n} - |\mathcal{K}|\right]}{0.8\, c'\sqrt{n}} \geq 0.25.$$

We now show that $e(U)$ is close to m with high probability. Note that $|m - \mathbb{E}[e(U)]| = O(n^{3/2})$, as $\mathbb{E}[e(U)] = e(U_0)/4$ and $|e(U_0) - 4m| = O(n^{3/2})$ from Lemma 3.5. Since the presence of edges in $G[U]$ are dependent only if they share a vertex, there are $O(n^3)$ dependent pairs of edges in $G[U]$. Thus we have $\mathrm{Var}(e(U)) \leq O(n^3) + \mathbb{E}[e(U)] = O(n^3)$. Since $|e(U) - m| \leq |e(U) - \mathbb{E}[e(U)]| + |\mathbb{E}[e(U)] - m|$, Chebyshev's inequality yields

$$\mathbb{P}\left[|e(U) - m| = O(n^{3/2})\right] \geq \mathbb{P}\left[|e(U) - \mathbb{E}[e(U)]| = O(n^{3/2})\right] \geq 1 - \frac{\mathrm{Var}(e(U))}{\omega(n^3)} \geq 0.9,$$

for large enough n. Fix an outcome U satisfying both events above, which hold together with probability at least $0.25 + 0.9 - 1 = 0.15$. Let $Q, \beta, M, \mathcal{I}_k$ be given by Lemma 3.7.

For each k, we have

$$\mathbb{E}[e_{k,0}] - \mathbb{E}[e_{k-1,0}] = \frac{1}{2}[e(\mathbf{Z}_{k,0}, U_0) - e(\mathbf{Z}_{k-1,0}, U_0)] + O(\sqrt{n}) = \Theta(n),$$

as $\mathbb{E}[d_U(\mathbf{x})] = d_{U_0}(\mathbf{x})/2 = \Theta(n)$ for $\mathbf{x} \in \mathbf{S}$. We find a subset from $\mathcal{K}$ whose realized values $e_{k,0}$ are well-separated by a linear gap. Specifically, let $\mathcal{K}' \subseteq \mathcal{K}$ contain every qth element of $\mathcal{K}$, for sufficiently large q such that the values of $\mathbb{E}[e_{k,0}]$ are separated by at least $4Qn$ for $k \in \mathcal{K}'$. Then, $|\mathcal{K}'| = \Theta(\sqrt{n})$ and Property 2 of Lemma 3.7 ensures that the values of $e_{k,0}$ are separated by at least $2Qn$ for $k \in \mathcal{K}'$.

We move on to partition the linear gaps between the values of $e_{k,0} : k \in \mathcal{K}'$ into square-root sized gaps. Let $k \in \mathcal{K}'$. By Property 3 of Lemma 3.7, we have $e_{k,c'\sqrt{n}} - e_{k,0} \geq 3\beta n$. Consider the range of integers between $e_{k,0}$ and $e_{k,0} + 3\beta n$. Within this range, we take $2\beta\sqrt{n}/M$ intervals of length $M\sqrt{n}$, each separated by a distance of $M\sqrt{n}/2 = \Omega(\sqrt{n})$. By Property 4 of Lemma 3.7, at most $\beta\sqrt{n}/M$ of these intervals contain no value of $e_{k,i}$. Consider the $\beta\sqrt{n}/M$ intervals that do contain a value of $e_{k,i}$ and pick a representative $e_{k,i}$ from each such interval. Let $\mathcal{I}'_k$ be the set of indices i corresponding to these representatives, and let $\mathcal{I}''_k = \mathcal{I}_k \cap \mathcal{I}'_k$. Then

$$|\mathcal{I}''_k| = |\mathcal{I}'_k| + |\mathcal{I}'_k| - |\mathcal{I}'_k \cup \mathcal{I}_k| \geq \beta\sqrt{n}/M - \beta c'\sqrt{n}/2M = \Omega(\sqrt{n})$$

and the values of $e_{k,i}$ are separated by $\Omega(\sqrt{n})$ for $i \in \mathcal{I}''_k$. Recall that $Q \geq 3\beta$, and so by construction $|e_{k,i} - e_{k,0}| \leq 3\beta n \leq Qn$ for $i \in \mathcal{I}''_k$. In particular, amongst choices of $k \in \mathcal{K}'$ and $\mathcal{I}''_k$, we already have $\Omega(n)$ distinct values of $e(U_{k,i})$ that are separated by $\Omega(\sqrt{n})$. It remains to fill the square-root sized gaps between these values.

In increasing order of these $e(U_{k,i})$, let $\mathcal{P}' \subseteq \{(k,i) : k \in \mathcal{K}', i \in \mathcal{I}''_k\}$ be the corresponding index set of every qth of these values. We enlarge q so that the values $e(U_{k,i})$ from $\mathcal{P}'$ are separated by at least $2Qn$. For each $(k,i) \in \mathcal{P}'$ we have $i \in \mathcal{I}_k$, so Property 1 of Lemma 3.7 furnishes a set $\mathbf{X}_{k,i}$ such that the values $d_{U_{k,i}}(\mathbf{x})$ are distinct for $\mathbf{x} \in \mathbf{X}_{k,i}$ while lying in a fixed interval of length $2Qn$ centered at $d/2$, where d is independent of k and i. In particular, for each $(k,i) \in \mathcal{P}'$, the set

$$\{e(U_{k,i} \cup \mathbf{x}_{k,i}) : \mathbf{x}_{k,i} \in \mathbf{X}_{k,i}\}$$

lie in an interval of length $2Qn$ centered at $e(U_{k,i}) + d/2$, and so these sets do not overlap among different $(k,i) \in \mathcal{P}'$.

Therefore,

$$|\{e(U \cup Z) : Z \subseteq W\}| \geq |\{e(U_{k,i} \cup \mathbf{x}_{k,i}) : (k,i) \in \mathcal{P}',\, \mathbf{x}_{k,i} \in \mathbf{X}_{k,i}\}| = \Omega(n^{3/2}).$$

4 The Erdős–McKay Conjecture

5 Concluding Remarks

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A Supplementary Proofs